



Figure 24 - Complexity forecast page

The complexity forecast page displays how complexity is predicted to evolve across different sectors, based on current sectorization. Users receive alerts for any potential hotspots and can choose which sector visualise turning on and off the sectors in the top left corner.

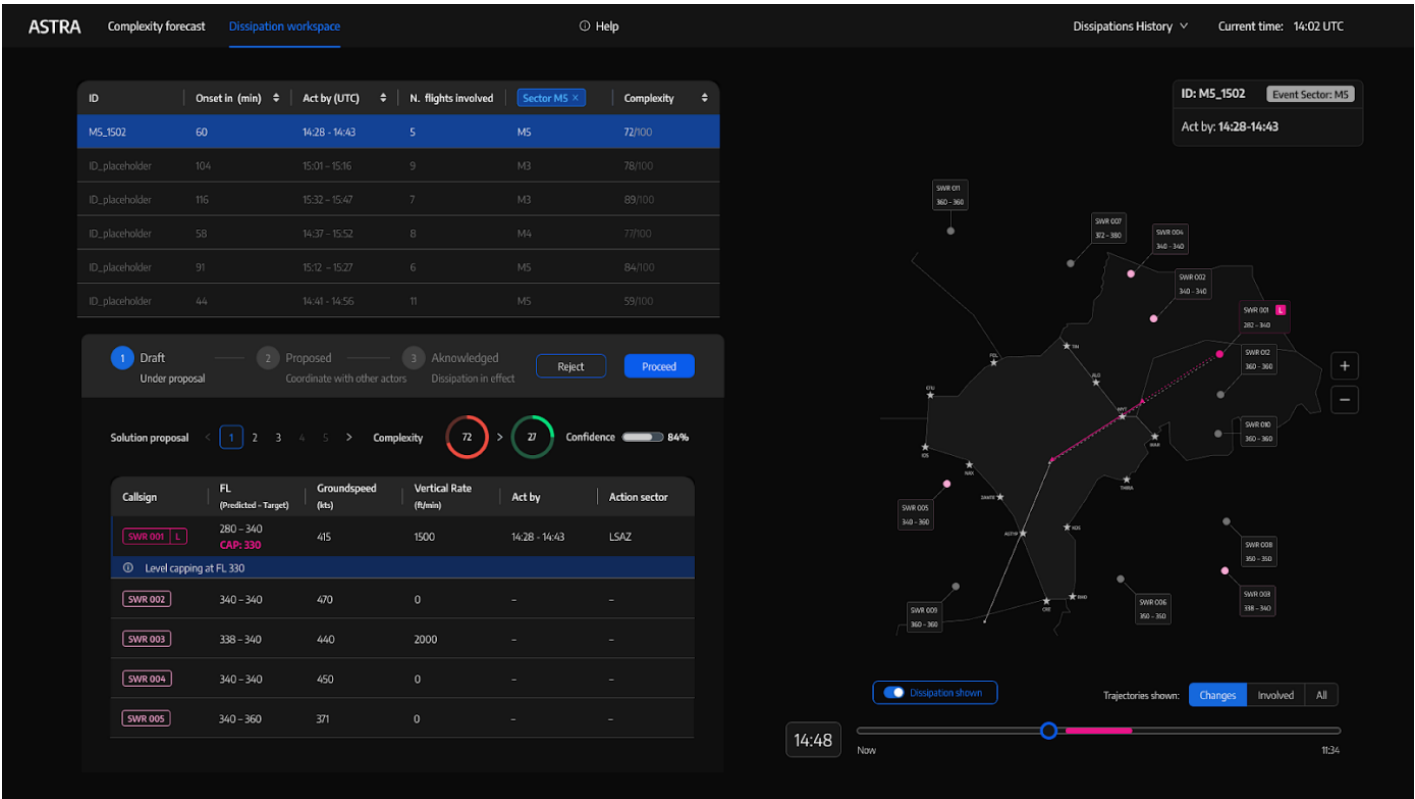


Figure 25 - Dissipation workspace

In this page, the users can see and analyse the details of each hotspot alert and explode the suggested dissipation strategies, step-by-step. Each element of the interface is explained in the images below.

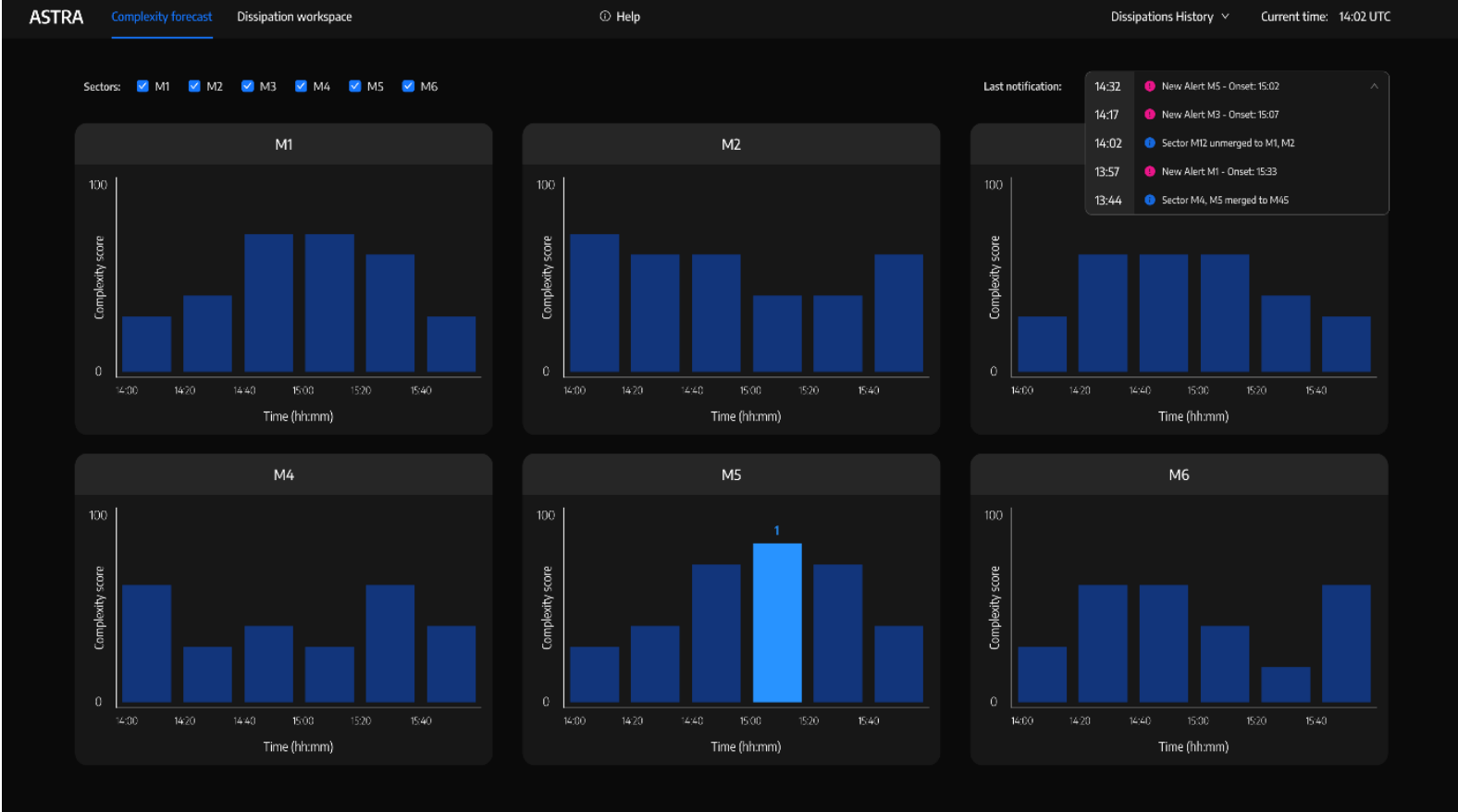


Figure 26 - Complexity forecast page with opened notification panel

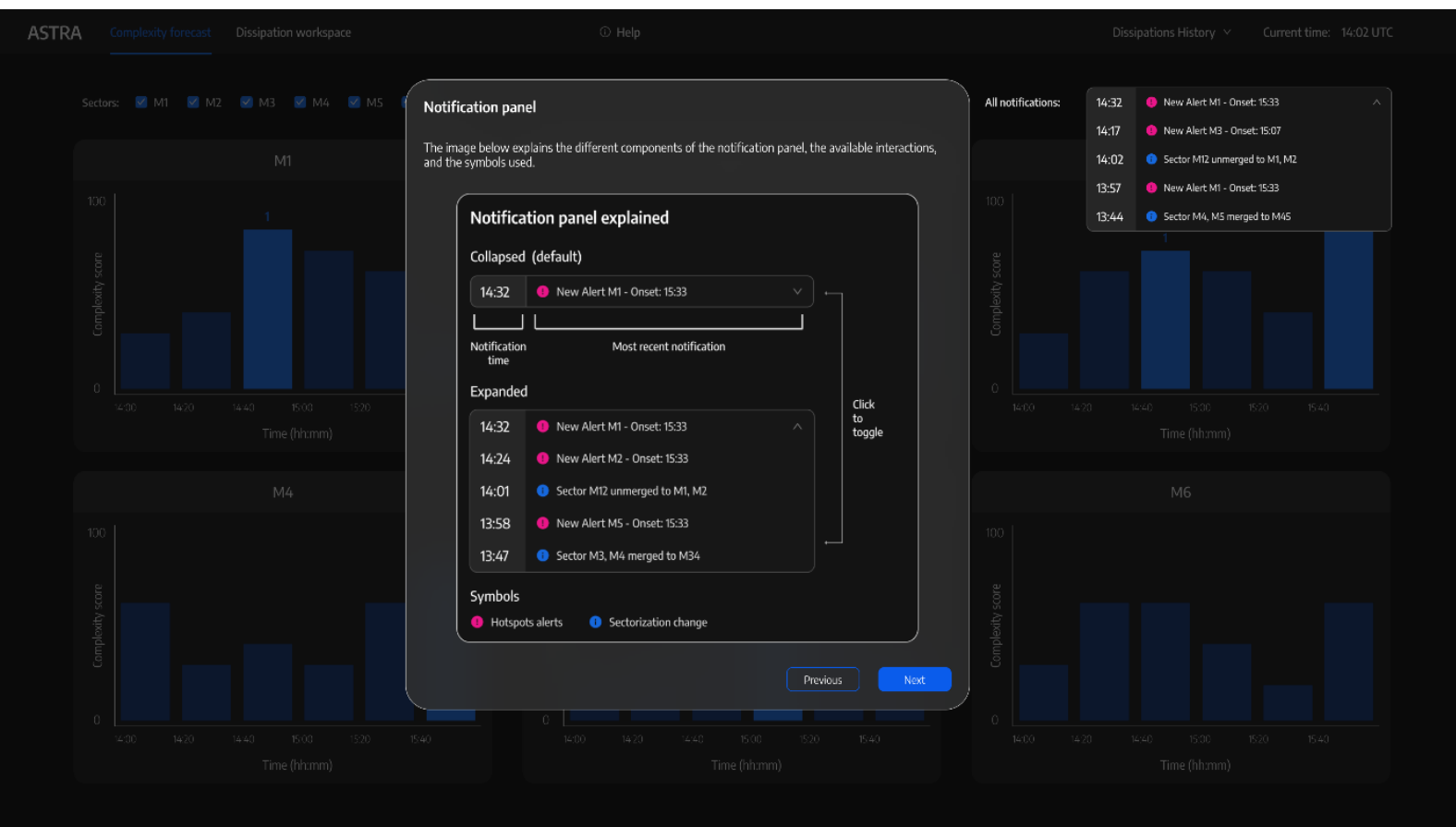


Figure 27 - Example of HMI walkthrough explanation process

In the figure, the notification panel is explained to the final users. Users can find two different types of events; (1) New hotspot alert, showing the sector and onset time; (2) Changes to airspace sectorisation (managed outside of ASTRA). By default, the notification panel is collapsed and shows only the latest notification.

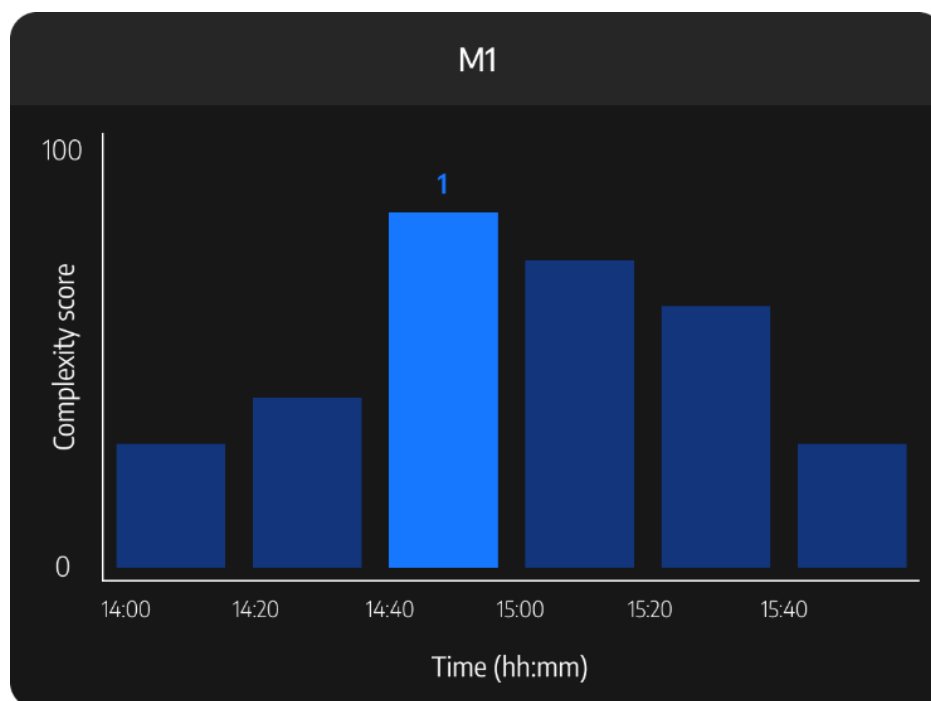


Figure 28 - Complexity forecast chart

The chart illustrates the predicted evolution of complexity in 20-minute intervals over the next two hours for each sector, based on the current sectorisation. Bars with hotspot alerts are highlighted, with the number of alerts shown above. By clicking on a highlighted bar, users access the Dissipations workspace, where the data is filtered to show details for that sector.

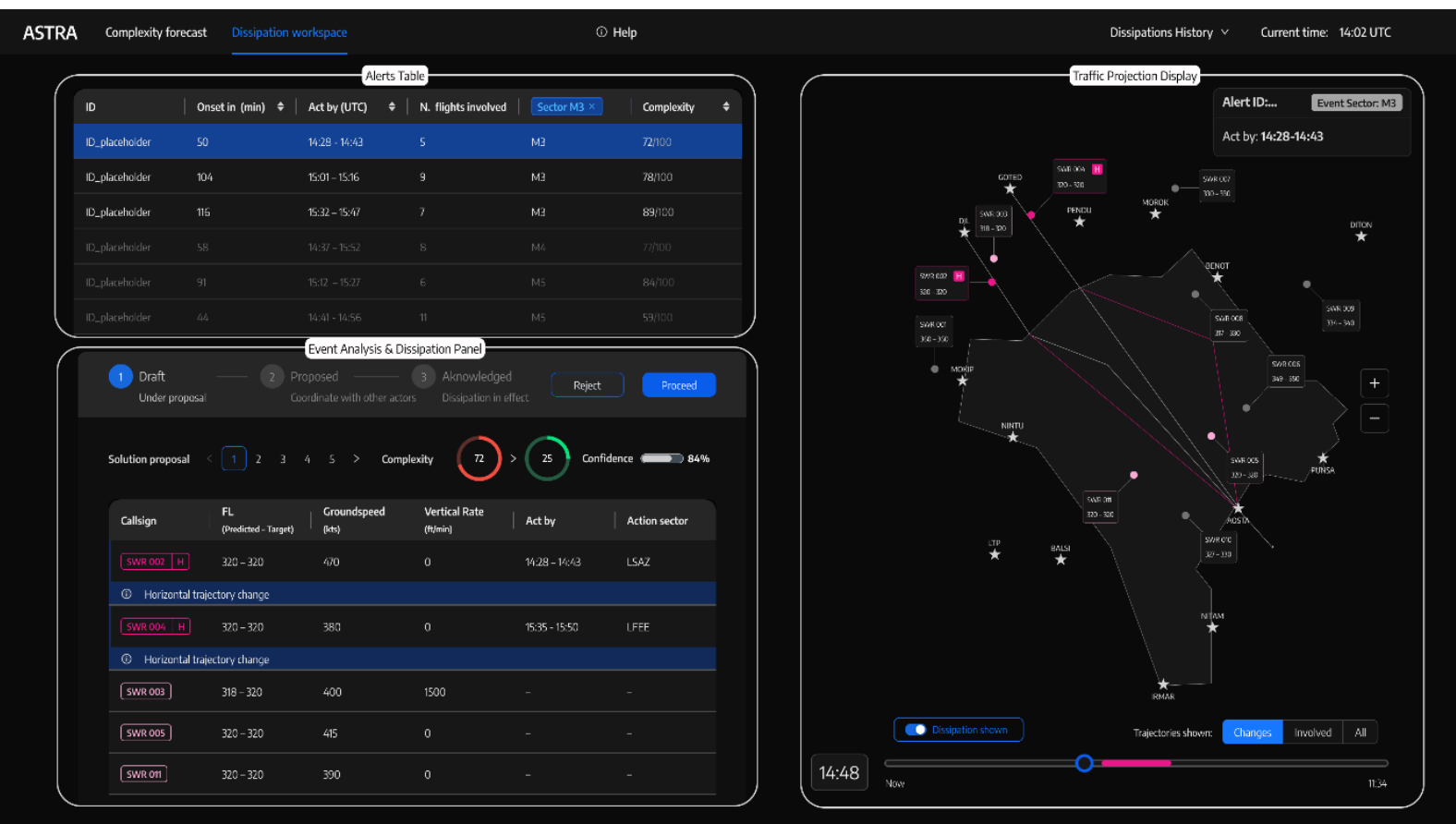


Figure 29 - Dissipation workspace in detail

The three main elements - alerts table, event analysis & dissipation table, Traffic prediction display - are highlighted. In the figure, an alert is selected in the Alerts table (the blue row). Details about the alert are shown in the Event Analysis & Dissipation Table and visually represented in the Traffic Projection Display.

ID	Onset in (min)	Act by (UTC)	N. flights involved	Sector M3 ×	Complexity
ID_placeholder	50	14:28 - 14:43	5	M3	72/100
ID_placeholder	104	15:01 - 15:16	9	M3	78/100
ID_placeholder	116	15:32 - 15:47	7	M3	89/100
ID_placeholder	58	14:37 - 15:52	8	M4	77/100
ID_placeholder	91	15:12 - 15:27	6	M5	84/100
ID_placeholder	44	14:41 - 14:56	11	M5	59/100

Figure 30 - Alerts table

The table displays the details of all received alerts. Some rows appear faded because a sector filter is applied, based on the alert the user clicked. Each column provides specific information, namely the ID, Onset In (the time - in minutes - until the hotspot is expected to begin), Act By (the time range within which pilots must start executing the dissipation instructions), the number of flights involved, the event sector (the sector where the hotspot is predicted to occur), the complexity.

1 Draft
Under proposal

2 Proposed
Coordinate with other actors

3 Acknowledged
Dissipation in effect

Reject

Proceed

Solution proposal

1

2

3

4

5

Complexity

72

25

Confidence 84%

Callsign	FL (Predicted - Target)	Groundspeed (kts)	Vertical Rate (ft/min)	Act by	Action sector
SWR 002 H	320 - 320	470	0	14:28 - 14:43	LSAZ
Horizontal trajectory change					
SWR 004 H	320 - 320	380	0	15:35 - 15:50	LFEE
Horizontal trajectory change					
SWR 003	318 - 320	400	1500	-	-
SWR 005	320 - 320	415	0	-	-
SWR 011	320 - 320	390	0	-	-

Figure 31 - Event analysis & Dissipation panel

In this panel the user can review the details of the selected alert and the suggested dissipation strategies by: (1) looking at details for each flight involved in the hotspot, including suggested trajectory modifications (inner table); (2) tracking the progress of dissipation (coordination with external actors is still handled outside of ASTRA); (3) exploring alternative dissipation strategies and visualize how each one affects overall complexity (top section).

Figure 32 - Traffic Projection Display

The user can visualize how traffic will evolve in the alert sector, both with and without the selected dissipation strategy. It is characterized by: (1) a map, designed like a radar display, featuring sector boundaries, reference waypoints, position symbols, labels, and trajectories; (2) flights, with labels and trajectories that follow a specific colour scheme; (3) a dissipation toggle, that allows users to switch between viewing how traffic evolves with or without the dissipation applied; (4) a trajectory view controls, that allows the user to choose which flight trajectories to display; (5) a timeslider, that allows

users to see how traffic evolves over time by dragging the time marker (visualizing the changes in traffic as time progresses). The duration of the hotspot is highlighted on the time slider, marking its start and end.

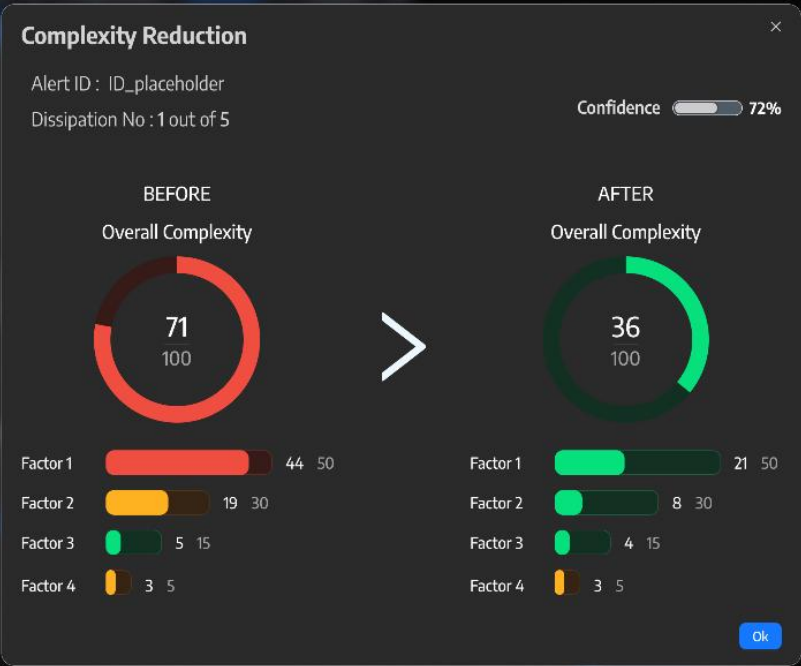


Figure 33 - Complexity reduction (XAI visualisation)

The overall complexity score ranges from a minimum to a maximum value (e.g., 0 to 100), combining all factors into a single metric. ASTRA also provides a confidence metric, shown as a percentage, indicating how certain ASTRA is that the proposed solution will successfully dissipate the hotspot.

ASTRA

Negotiation and Coordination steps outside ASTRA

Here is a list of hypothetical actions and events required to coordinate the dissipation with the other ACC, assuming the dissipation is accepted:

Actors involved	Action Items
- FMP at Event ACC (You): FMP-E - Supervisor at Event ACC: Sup-E - FMP at Acting Centre: FMP-A - Planner ATCO at Acting Sector: Planner-A - Executive ATCO at Acting Sector: Executive-A	FMP-E (You) consults Sup-E - Sup-E gives go-ahead FMP-E (You) contacts FMP-A and coordinates (by phone) with this position - FMP-A acknowledges - FMP-A communicates with Planner-A - Planner-A accepts - Planner-A coordinates with Executive-A - Executive-A acts on flight 1 - FMP-A inputs message on FPL of flight concerned level capping condition in RMK field FMP-E (You) informs upstream ACC about FL deviation for this flight.

When ready, click on **Next to proceed** ➡

Next

Figure 35 - Coordination steps presented to the user

When the main user of ASTRA considers the solution as acceptable, they need to coordinate with colleagues from the Action ACC. They have to discuss and negotiate the solution suggested by ASTRA. For now, all these steps must be carried out outside ASTRA using the conventional communication methods between ACCs (e.g., telephone).

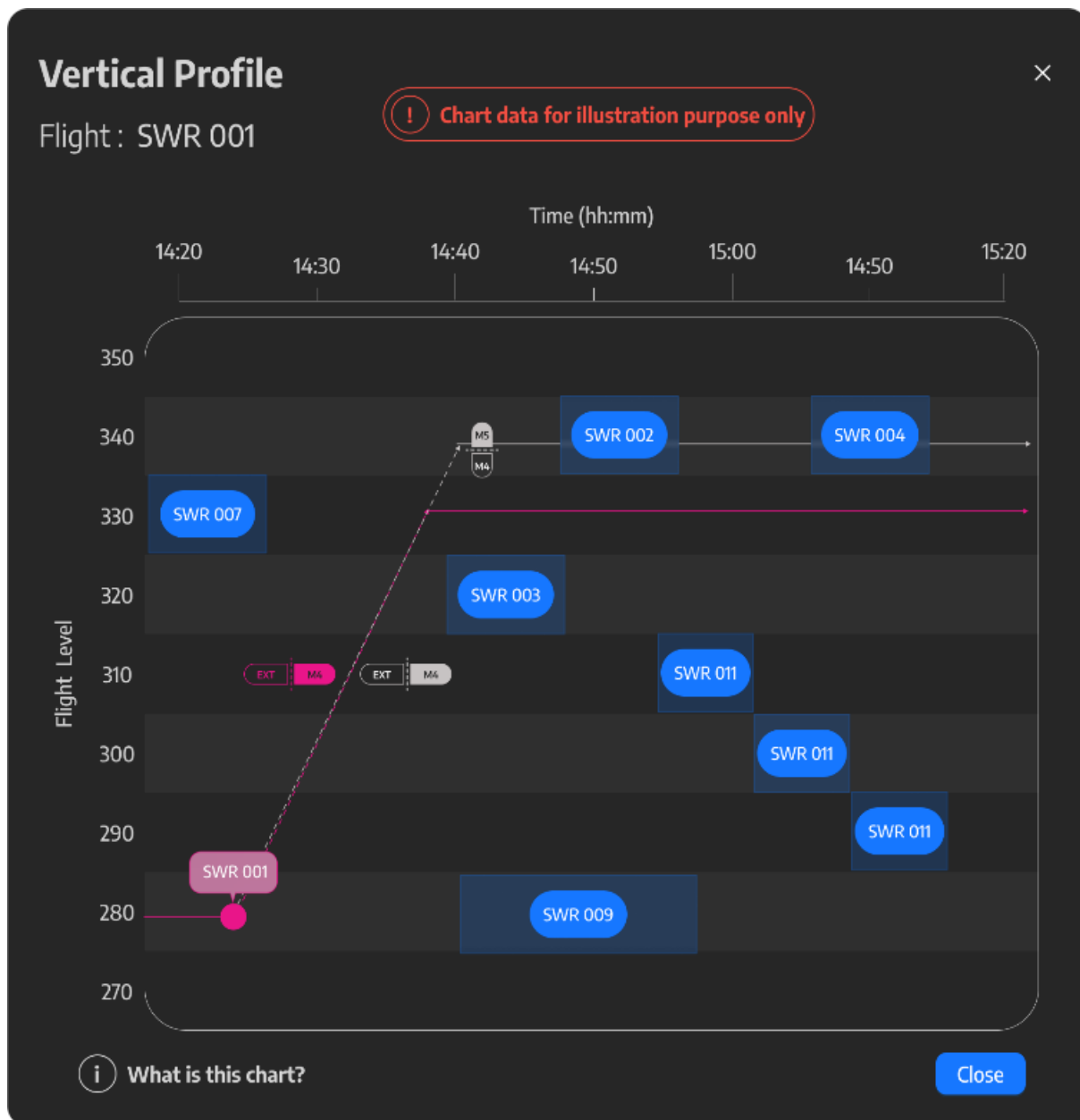


Figure 34 - Vertical profile

This chart shows the vertical trajectory of the selected flight, comparing its initial and modified trajectories over time. It is a "what-if" tool that helps you visualize potential proximities with other aircraft and sector transitions. Blue rectangles represent no-go zones. If the selected flight crosses the flight level during the time interval covered by the blue block, it will enter a proximity event with another aircraft—either less than [XX] NM horizontally or [YYYY] ft vertically. Sector transition markers indicate when and where the flight crosses sector boundaries, for both the initial and modified trajectories.